

Acids and bases

ACIDS AND BASES ARE CHEMICAL OPPOSITES, BUT THESE TWO TYPES OF COMPOUNDS ARE CLOSELY RELATED.

The chemistry of acids and bases is driven by hydrogen ions. Acids are substances that produce hydrogen ions; bases are substances that react with acids by accepting these hydrogen ions.

What is an acid?

Acids are compounds that release positively charged particles of hydrogen, called hydrogen ions (H^+), when dissolved in water. These ions are highly reactive and can bond to other substances and have a corrosive effect on them. The strength of an acid depends on the number of hydrogen ions that it can release.

▽ Strong acids

The most powerful acids are ionically bonded compounds (see page 112). They split into hydrogen and other ions completely when dissolved in water, thus releasing large quantities of free hydrogen ions.

Name	Formula	Where it is found
hydrochloric acid	HCl	the stomach
sulfuric acid	H_2SO_4	car batteries
nitric acid	HNO_3	process to make fertilizers

What is a base?

A base is a compound that reacts with an acid by accepting its hydrogen ions. The most reactive bases are soluble compounds called alkalis. As it dissolves, an alkali releases hydroxide ions (OH^-). The hydrogen and hydroxide ions combine very readily to form water, so the reaction between an acid and an alkali is often vigorous.

Neutralization

The reaction between an acid and an alkali (or other base, such as an oxide) is called neutralization, because it results in products that are neither acid nor alkali. One of the products is always water. The other, known as the salt, is a compound formed from the left-over ions.

SEE ALSO

- ◀ 89 Acid rain
- ◀ 112–113 Ionic bonding
- ◀ 114–115 Covalent bonding
- ◀ 120 Why alkali?
- ◀ 136–137 Rates of reaction

DNA, the chemical that carries genetic code, is a type of acid.

▽ Weak acids

Acids which have a covalent structure (see page 114) do not break up into ions so easily. They have complex molecules, but sometimes the bond holding a certain hydrogen ion weakens, allowing it to break off.

Name	Formula	Where it is found
citric acid	$C_6H_8O_7$	lemon juice
ethanoic acid	CH_3COOH	vinegar
formic acid	$HCOOH$	ant stings

▽ Common alkalis

Any compound with a hydroxide ion is known as an alkali. Alkalis are used in industry to make soaps and are added to waste to help it decay more rapidly.

Name	Formula	Where it is found
sodium hydroxide	NaOH	oven cleaner
magnesium hydroxide	$Mg(OH)_2$	indigestion tablets
potassium hydroxide	KOH	soap

▽ General equation

An acid and base always react to produce a salt and water. The salt produced when hydrochloric acid (HCl) reacts with sodium hydroxide (NaOH) is sodium chloride (NaCl)— better known as common salt.

